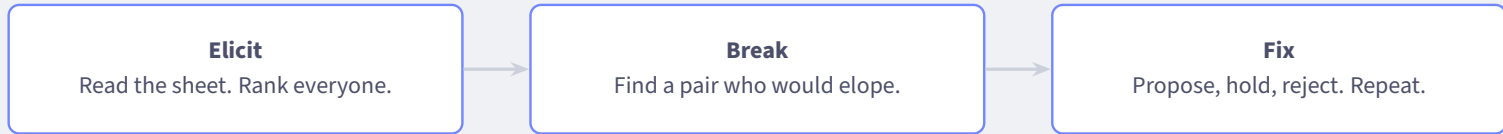


Matching games

Four mathematicians, four physicists, one stable matching



Mathematicians

Gauss				
Noether				
Turing				
Euler				

Physicists

Einstein				
Curie				
Newton				
Feynman				

Most preferred first. These come from the preference sheet, not from me.

Pair everyone up however you like:

with

with

with

with

Now find two people who would both rather have each other. That pair is a _____, and the matching is _____.

.....

.....

.....

.....

.....

Round 1

Gauss

Einstein

Noether

Curie

Turing

Newton

Euler

Feynman

Round 2

Gauss

Einstein

Noether

Curie

Turing

Newton

Euler

Feynman

Draw each proposal as an arrow. Circle the one held, and cross out the rejections.

Round 3

Gauss

Einstein

Noether

Curie

Turing

Newton

Euler

Feynman

Round 4

Gauss

Einstein

Noether

Curie

Turing

Newton

Euler

Feynman

Stop when nobody is left holding a proposal they have not accepted.

	with			with	
	with			with	

Check it: take any pair who are *not* matched to each other, and show at least one of them prefers who they have.

.....

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.....

Who does better out of this, the proposers or the proposed to?

What we just did in the room is **Question 1** on the Matching Games page, with three on each side instead of four.

- (a) Define a blocking pair and a stable matching, and write the preferences as maps. [5]
- (b) Run Gale-Shapley with the mathematicians proposing, showing each step. [7]
- (c) Write down the resulting matching. [3]
- (d) Verify it is stable by checking for a blocking pair. [4]
- (e) Whose optimum is it, and why can the proposing side never do better? [6]

vknight.org/gt/topics/matching-games.html